

Domain and range

Two questions, asked in a fixed order: what may go in, and what then comes out.

LESSON 11–12

2 × 40 MIN

ALGEBRA 10, §1.3

P1 · 2.2

LESSON CLOCK

6'

14'

14'

12'

8'

6'

The two questions	6 min
Finding a domain — three traps	14 min
Finding a range	14 min
Restricted domains	12 min
Practice	8 min
Homework	6 min

LEARNING OBJECTIVES

- Find the natural domain of a rational, root and polynomial function.
- Find the range of a linear, quadratic and simple rational function.
- Write domain and range in set notation and in interval notation.
- Explain the effect of a restricted domain on the range.

TEXTBOOK

National	pp. 35–41
Cambridge	Pure Mathematics 1, pp. 30–36
Workbook	P1 Exercise 2B

01

Explanation

Domain first, always

THE ORDER IS NOT OPTIONAL

The **domain** is the set of inputs the rule accepts. The **range** is the set of outputs it then produces. You cannot know the range until you know which inputs were allowed — so find the domain first, every time.

When a function is given by a formula and nothing else, its domain is the largest set of real numbers the formula makes sense on. That is the **natural domain**.

Three things a formula can refuse

REFUSAL	CONDITION TO IMPOSE	EXAMPLE	DOMAIN
division by zero	denominator $\neq 0$	$\frac{1}{x-3}$	$x \neq 3$
even root of a negative	radicand ≥ 0	$\sqrt{x-2}$	$x \geq 2$
both at once	radicand > 0	$\frac{1}{\sqrt{x-2}}$	$x > 2$

A polynomial refuses nothing: its natural domain is always $\mathbb{R}$.

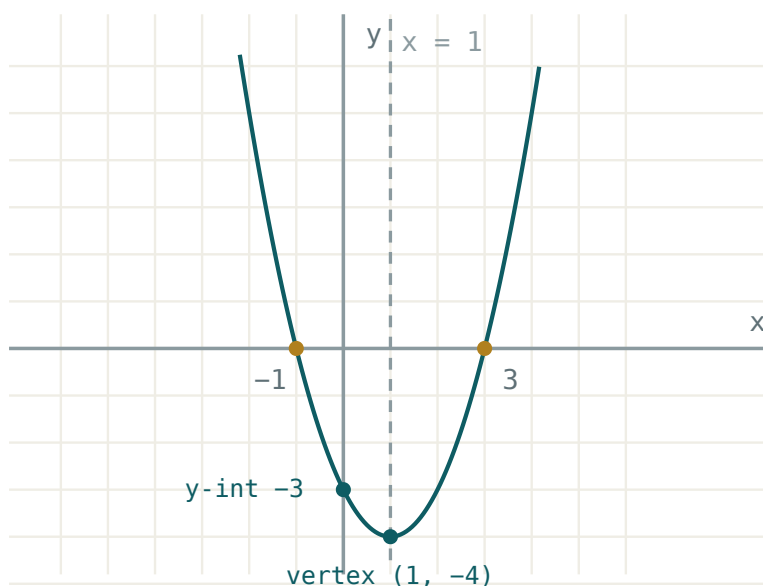
CANCEL AFTER THE DOMAIN, NOT BEFORE

$f(x) = \frac{x^2 - 9}{x - 3}$ simplifies to $x + 3$, but the domain is still $x \neq 3$. The graph is the line with a hole punched at $(3, 6)$. Cancelling never restores a forbidden input.

Finding the range

Four reliable methods, in the order you should try them:

1. **Linear** — the range is all of $\mathbb{R}$ unless the domain is restricted.
2. **Quadratic** — complete the square; the vertex gives the least (or greatest) value.
3. **Rational** — look for the value the output can never take: the horizontal asymptote.
4. **Anything** — sketch it and read the y -values the curve covers.



The range of a parabola starts at the vertex. Read it off the y -axis, not the x -axis.

$$y = x^2 - 6x + 11 = (x - 3)^2 + 2 \Rightarrow \text{range } y \geq 2$$

For $f(x) = \frac{1}{x}$ the output 0 is impossible, because $1 \div anything$ is never 0 . So the range is $y \neq 0$ — the same shape of answer as the domain, one value removed.

Restricting the domain

Give a function a smaller domain and the range shrinks with it. For $f(x) = x^2$:

DOMAIN	RANGE	KIND
$\mathbb{R}$	$y \geq 0$	many-to-one
$x \geq 0$	$y \geq 0$	one-to-one
$1 \leq x \leq 3$	$1 \leq y \leq 9$	one-to-one
$-1 \leq x \leq 3$	$0 \leq y \leq 9$	many-to-one

THE LAST ROW IS THE TRAP

On $-1 \leq x \leq 3$ the smallest output is **not** $f(-1) = 1$. The vertex $x = 0$ lies inside the interval, so the least value is 0 . Always check whether the vertex is inside the restricted domain before reading the endpoints.

02

Worked examples

EXAMPLE 1 Find the domain and range of $f(x) = \sqrt{x + 4}$.

1

Radicand ≥ 0 .
 $x + 4 \geq 0 \Rightarrow x \geq -4$

2

A square root gives non-negative outputs.
 $f(x) \geq 0$

3

At $x = -4$ the output is 0 and it grows without bound.

ANSWER

domain $x \geq -4$, range $y \geq 0$

EXAMPLE 2 Find the domain and range of $f(x) = \frac{2x}{x-1}$.

① Denominator $\neq 0$.

$$x \neq 1$$

② Rearrange for x : $y(x-1) = 2x$

$$yx - y = 2x$$

③ $x(y-2) = y \Rightarrow x = \frac{y}{y-2}$

This fails when $y = 2$.

④ So 2 is the one unreachable output.

ANSWER

domain $x \neq 1$, range $y \neq 2$

EXAMPLE 3 Find the range of $f(x) = x^2 - 4x + 7$ on $0 \leq x \leq 3$.

① Complete the square.

$$(x-2)^2 + 3$$

② The vertex $x = 2$ lies inside $[0, 3]$.

Least value 3.

③ Test both ends.

$$f(0) = 7, f(3) = 4$$

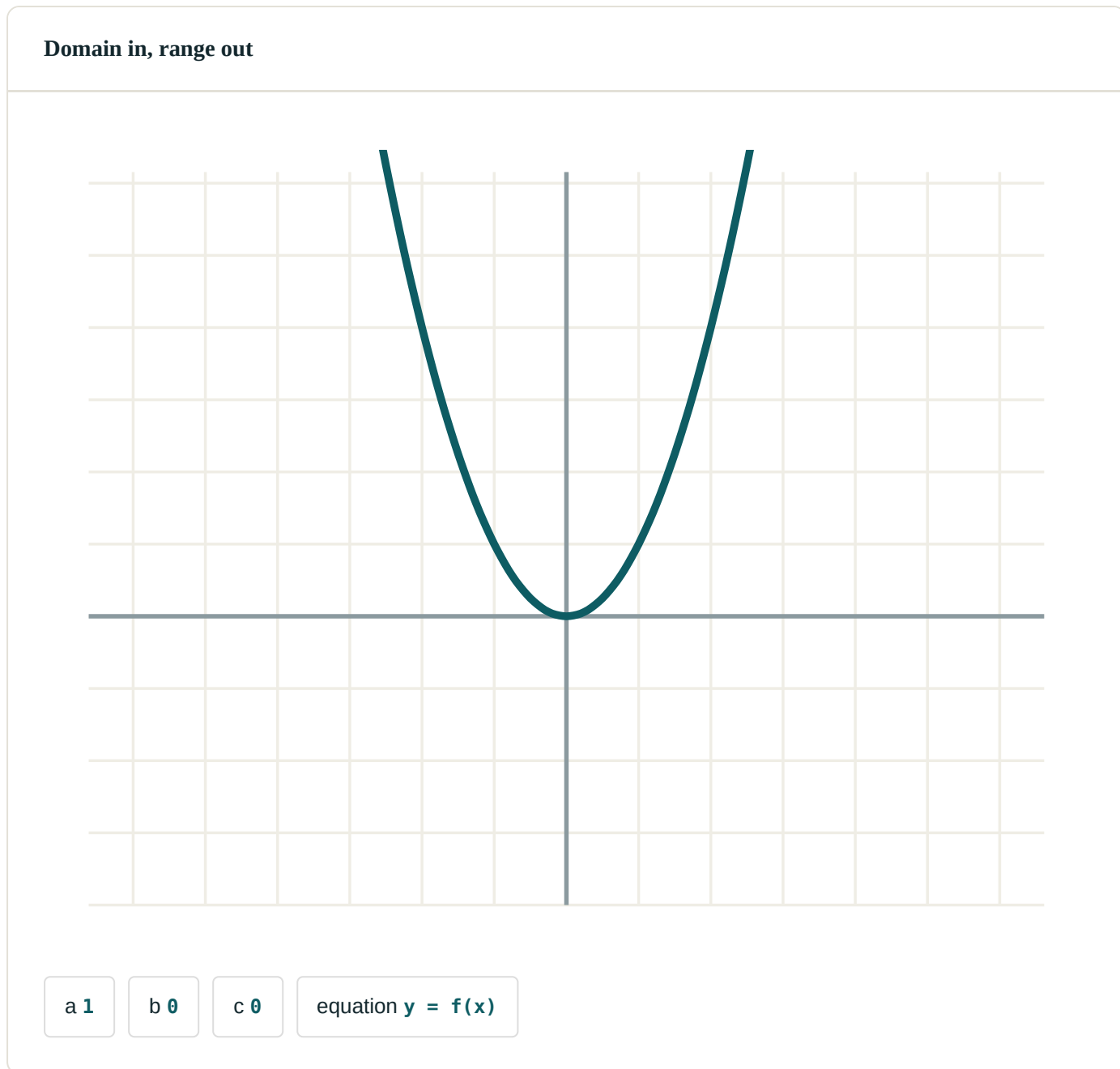
④ Greatest is 7.

ANSWER

$$3 \leq y \leq 7$$

03 Interactive model

Drag the ends of the domain and watch the range respond.

**04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Domain	Aniqlanish sohasi	Область определения
Range	Qiymatlar sohasi	Область значений
Natural domain	Tabiiy aniqlanish sohasi	Естественная область определения
Interval notation	Oraliq belgilanishi	Интервальная запись
Excluded value	Chetlatilgan qiymat	Исключённое значение
Restriction	Cheklash	Ограничение
Asymptote	Asimptota	Асимптота
Minimum value	Eng kichik qiymat	Наименьшее значение
Set notation	To'plam belgilanishi	Обозначение множеств

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The natural domain of $\frac{1}{x+5}$ is:

$x \neq 5$

$x \neq -5$

$x \geq -5$

$\mathbb{R}$

The range of $y = x^2 + 3$ is:

$y \geq 0$

$y \geq 3$

$y > 3$

$\mathbb{R}$

The domain of $\sqrt{5-x}$ is:

$x \geq 5$

$x \leq 5$

$x \neq 5$

$\mathbb{R}$

The range of any non-constant linear function on $\mathbb{R}$ is:

$$y \geq 0$$

$$\mathbb{R}$$

$$y \neq 0$$

one value

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $f(x) = 3x - 1$

ANSWER $\mathbb{R}$

2. Domain of $\frac{1}{x}$

ANSWER $x \neq 0$

3. Domain of $\sqrt{x}$

ANSWER $x \geq 0$

4. Range of $y = x^2$

ANSWER $y \geq 0$

5. Range of $y = x^2 + 5$

ANSWER $y \geq 5$

6. Domain of $\frac{1}{x - 7}$

ANSWER $x \neq 7$

7. Range of $y = 4$

ANSWER $\{4\}$

Medium · the standard question

7 PROBLEMS

1. Domain of $\sqrt{2x - 6}$

ANSWER $x \geq 3$

2. Domain of $\frac{x}{x^2 - 4}$

ANSWER $x \neq \pm 2$

3. Range of $y = (x - 1)^2 - 6$

ANSWER $y \geq -6$

4. Range of $y = 7 - x^2$

ANSWER $y \leq 7$

5. Range of $y = \frac{1}{x} + 3$

ANSWER $y \neq 3$

6. Range of $y = 2x + 1$ on $0 \leq x \leq 4$

ANSWER $1 \leq y \leq 9$

7. Domain of $\frac{1}{\sqrt{x - 1}}$

ANSWER $x > 1$

Hard · stretch and reason

7 PROBLEMS

1. Range of $y = x^2 - 6x + 5$ on $0 \leq x \leq 2$

ANSWER $-3 \leq y \leq 5$

2. Range of $y = x^2 - 6x + 5$ on $1 \leq x \leq 5$

ANSWER $-4 \leq y \leq 0$

3. Range of $y = \frac{3x + 1}{x - 2}$

ANSWER $y \neq 3$

4. Domain of $\sqrt{x^2 - 9}$

ANSWER $x \leq -3$ or $x \geq 3$

5. Domain of $\sqrt{6 + x - x^2}$

ANSWER $-2 \leq x \leq 3$

6. Find k so $y = x^2 + kx + 10$ has range $y \geq 1$

ANSWER $k = \pm 6$

7. Why does $\frac{x^2 - 1}{x - 1}$ have range $y \neq 2$?

ANSWER It is $x + 1$ with the point at $x = 1$ removed

07 Homework

Homework — 5 tasks

Every answer must say *domain* and *range* separately, in that order.

- Domain and range of $f(x) = \sqrt{x - 3}$.
- Domain and range of $f(x) = \frac{5}{x + 2}$.
- Range of $f(x) = x^2 - 8x + 20$ on $\mathbb{R}$, then on $0 \leq x \leq 3$.

4. Explain why $\frac{x^2 - 4}{x + 2}$ does not have domain $\mathbb{R}$.
5. Invent a function with domain $x > 1$ and range $y > 0$.